

Erdős Problem #1117

EP-1117

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f(z)$ be an entire function which is not a monomial. Let $u(r)$ count the number of z with $|z| = r$ such that $|f(z)| = \max_{|z|=r} |f(z)|$. (This is a finite quantity if f is not a monomial.)

Is it possible for $\limsup u(r) = \infty$? Is it possible for $\liminf u(r) = \infty$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>